Spatial Syntax of the Salt Police Station

This project converts the Salt Police Station into a series of mathematical graphs using TopologicPy. Each graph answers a specific question about how the building enforces spatial hierarchy, who can move where, who can see what, and who is isolated from whom.

Spatial Zone Color Coding

- Public: Lobby, entry spaces
- Private: Staff offices, admin
- Circulation: Corridors, spine, stairs
- Cells: Detainee holding spaces

01 Building Geometry

Extruded room volumes colored by spatial zone. The red circulation spine bisects the plan. Detainee cells (black) are peripheral. This is the baseline — geometry only, no relational data.

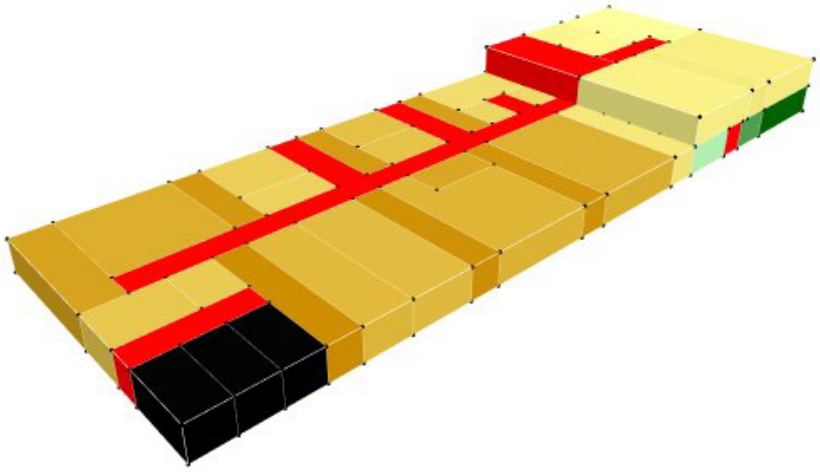

Fig. 1: 3D volumes colored by zone.

02 Primal Graph: Geometric Adjacency

Nodes = rooms · Edges = any shared face (wall contact), regardless of doors. Node size encodes **degree** — rooms bordering more spaces appear larger, revealing the geometric backbone. Does not distinguish between a wall and a door.

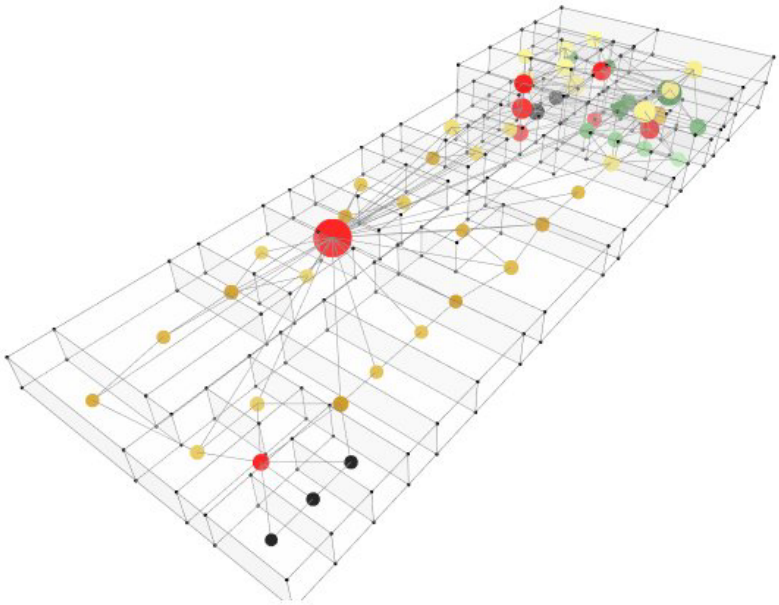

Fig. 2: Primal graph. Large nodes = high geometric connectivity.

03 Movement Graph: Physical Access

Edges form **only where a door or glass door exists**. Each aperture becomes its own node sitting between the rooms it connects. This graph enforces the security protocol topologically.

- Solid door: physical passage only
- Glass door: passage + visual connection

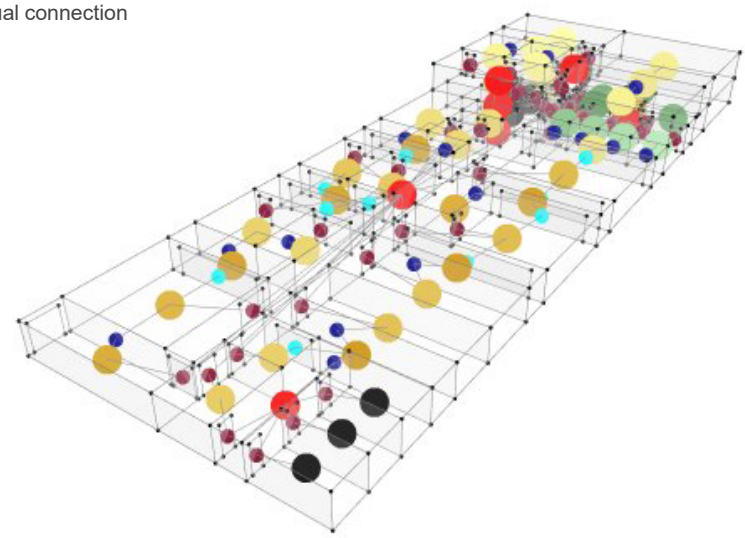

Fig. 3: Movement graph. Aperture nodes sit at wall thresholds. Edge = door exists.

04 — Visual Graph: Line of Sight

Edges form through **glass doors and windows only**, no physical movement, only visual connection.
Tests whether internal patios and glass thresholds create a surveillance network without requiring physical patrol.

- Glass door: visual + physical
- Window: visual only

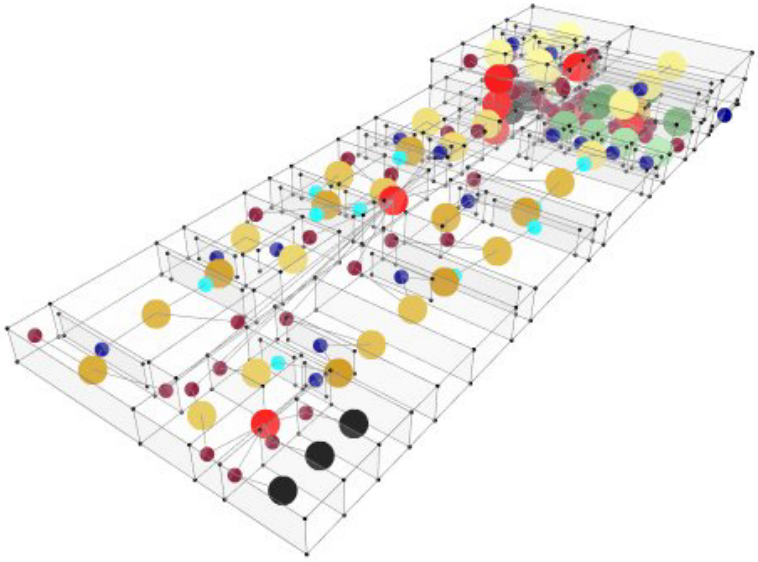

Fig. 4: Visual graph. Dense aperture connections = high surveillance reach.

Future Graphs to Explore

Detainee Escape Distance	Shortest path (door count) from each cell to every exit. Maps topological distance as a security metric.
Betweenness Centrality	Identifies architectural chokepoints, spaces all shortest paths pass through. High-betweenness nodes should be staffed.
Visibility Reach	From each node, count rooms reachable within N visual hops. Locates optimal guard positions.
Zone Permeability	Counts movement edges crossing security zone boundaries. Direct public-to-cell edges are security flags.
Geometry vs. Access Delta	Subtracts movement graph from primal graph. Remaining edges are intentional barriers, the security logic made visible.